SANIYA SARATKAR

DATA SCIENTIST • AI / ML ENGINEER

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Professional Summary: Early-career Data Scientist and AI/ML practitioner with experience in Python, machine learning, statistical analysis, NLP, forecasting, computer vision and Generative AI. Worked on customer intelligence, ESG research, satellite data and applied AI projects. Awarded an $8,500 UC Berkeley AI Scholarship.

CORE TECHNICAL SKILLS

Machine Learning & AI: Scikit-learn, Regression, Classification, Random Forest, XGBoost, CNN, LSTM, Natural Language Processing (NLP), Sentiment Analysis, Feature Engineering, Model Auditing.
Time-Series & Forecasting: ARIMA, SARIMA, Chronos Foundation Models, Statistical Testing (ADF, stationarity), Econometrics.
Generative AI & Vision: Large Language Models (LLMs), RAG, Prompt Engineering, FastAPI, Qdrant Vector DB, Ollama, MediaPipe, OpenCV.
Languages & Platforms: Python (Pandas, NumPy, PyTorch), SQL, MySQL, Google Cloud Platform, Google Earth Engine, Docker, Git.

PROFESSIONAL INTERNSHIP & RESEARCH EXPERIENCE

Deeptimaan Systems — *Data Science Intern* Feb 2025 - Jan 2026 | Hyderabad, India

- Developed machine learning sentiment-analysis workflows using approximately 50,000 online social media records.
- Worked on CRM customer intelligence, NLP text preprocessing, and automated reporting pipelines via n8n and Docker (21.25% effort saved).

IIT Kharagpur — *Research Analyst Intern* Mar 2025 - Aug 2025 | Kharagpur, India

- Analysed ~100,000 records across 14 years and 50 companies, exploring forecasting workflows using ARIMA, SARIMA, LSTM, and Chronos.
- Investigated lead-lag correlations between corporate governance scores and downside volatility, contributing to research publication.

ISRO – National Remote Sensing Centre (NRSC) — *Project Intern* May 2025 - Oct 2025 | Hyderabad, India

- Analysed 200,000+ satellite and geospatial records using Python, Google Earth Engine (GEE), and Copernicus Sentinel datasets.
- Developed an analytical visualization prototype mapping longitudinal environmental patterns; presented to senior ISRO scientists.

UC Berkeley (Applied AI Intern, *$8,500 AI Scholar*, Jul 2025) • Ignite Center (AI Mentor, Dec 2025 - Apr 2026) USA / Remote

- Explored applied AI, LLM evaluation pipelines, and mentored student researchers on ESG sustainability predictive modeling.

KEY PROJECT CASE STUDIES

SignAssistant: Indian Sign Language Translation Engine | *Python, MediaPipe, PyTorch, Scikit-learn, Streamlit*
- Engineered real-time gesture-to-speech system with 226-dim geometric invariants, achieving **99.61% DL accuracy** and 31ms edge CPU latency.

MediEvidence: Retrieval-Augmented Clinical Evidence Assistant | *Python, FastAPI, Qdrant Vector DB, Ollama, RAG*
- Built domain-specific clinical RAG assistant retrieving biomedical literature with verified PubMed citations and out-of-domain guardrails.

BacMotionAI: Bacterial Motility Under Chemical Stimuli | *Python, Scikit-learn, Random Forest, Gradient Boosting*
- Classified 2,400 swimming trajectories across 4 chemical conditions; Random Forest achieved **87.3% test accuracy** and 0.977 ROC-AUC.

ESG Indicators & Financial Market Forecasting | *Python, Pandas, ARIMA, SARIMA, LSTM, Chronos*
- Benchmarked classical econometric models and deep learning foundation models on 14 years of longitudinal market data.

EDUCATION

B.Tech in Artificial Intelligence & Data Science
Adani Institute of Higher Education & Research (AIHER), Nagpur
Duration: 2022 – 2026 | **CGPA:** 8.35 / 10.0

CERTIFICATIONS & LEADERSHIP

- **Google Data Analytics** Professional Certificate
- **$8,500 UC Berkeley AI Scholarship** Recipient
- **20+ Scholarly Publications** on Google Scholar
- **GDG Data Science Head • Student Council VP** (50+ events)